Numerical systems and analysis

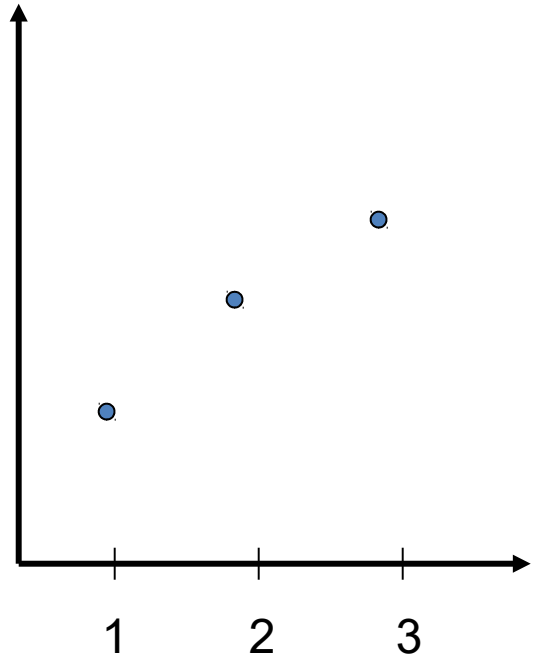
CHAPTER (4) CURVE FITTING: LEAST SQUARE REGRESSION

Chapter 17 of the textbook

Motivation

- Given a set of experimental data:

x	1	2	3
y	5.1	5.9	6.3



The relationship between ***x*** and ***y*** may not be clear.

Find a function ***f(x)*** that best fit the data

Motivation..

- In engineering, two types of applications are encountered:

1. **Trend analysis:** Predicting values of dependent variable, may include extrapolation beyond data points or interpolation between data points.
2. **Hypothesis testing:** Comparing existing mathematical model with measured data.

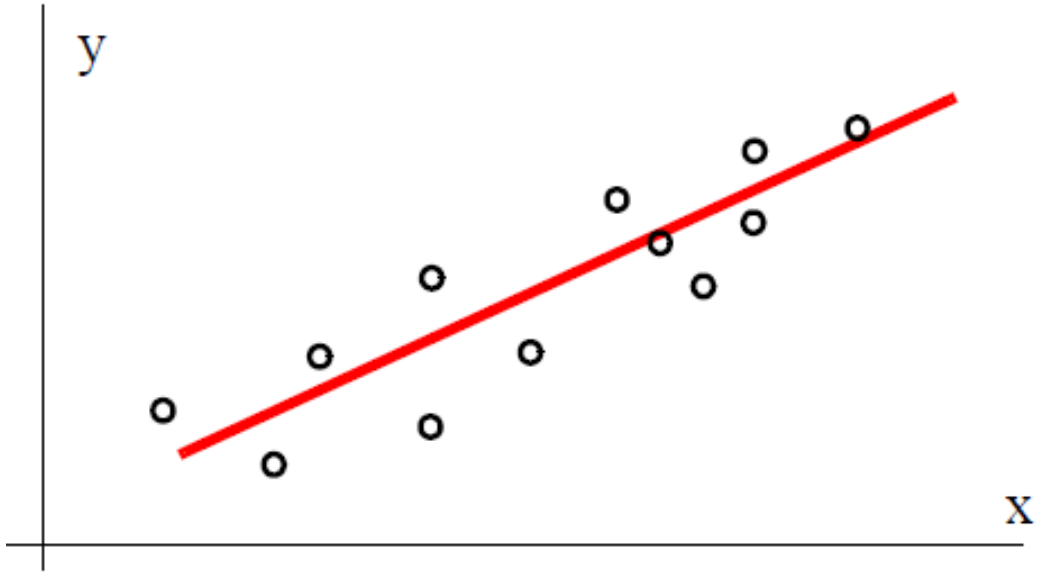
- What is the best mathematical function *f* that represents the dataset?
- What is the best criterion to assess the fitting of the function *f* to the data?

Curve Fitting

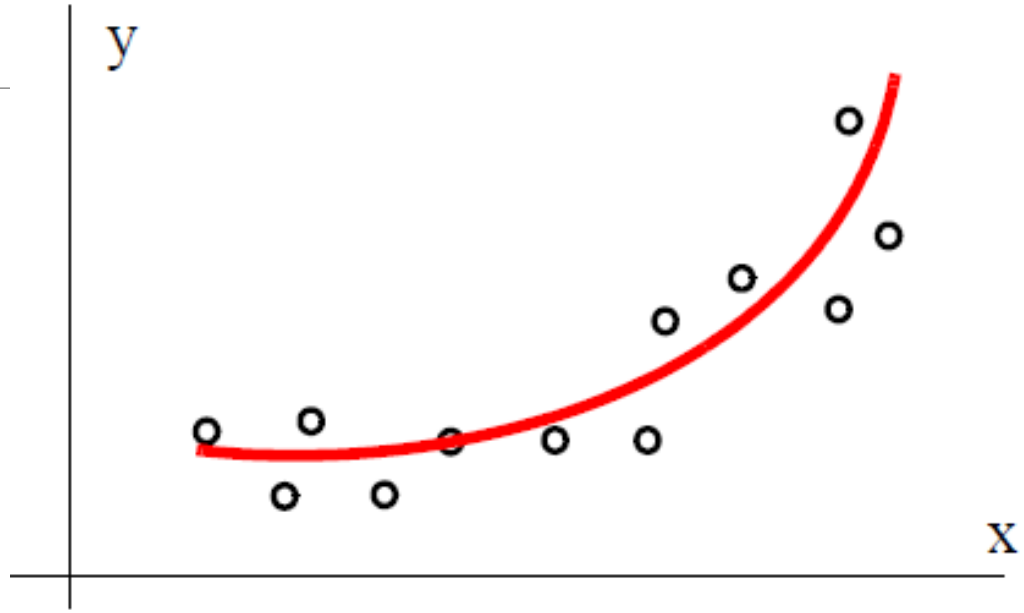
- Given a set of tabulated data, find a curve or a function that *best represents the data*.

- Given:
  - ✓ The tabulated **data**
  - ✓ The **form** of the function
  - ✓ The curve fitting **criteria**

- Find the **unknown** coefficients



Linear regression



Nonlinear regression

Outline

- Linear regression
- Nonlinear models using least squares
  - Exponential model
  - Simple power model
  - Saturation-growth-rate model
- Polynomial Regression
- Nonlinear Regression (Gauss-Newton method)

(1) Linear Regression

Linear Regression

- The purpose of a least-squares approximation is fitting a straight line to a set of points (observations):

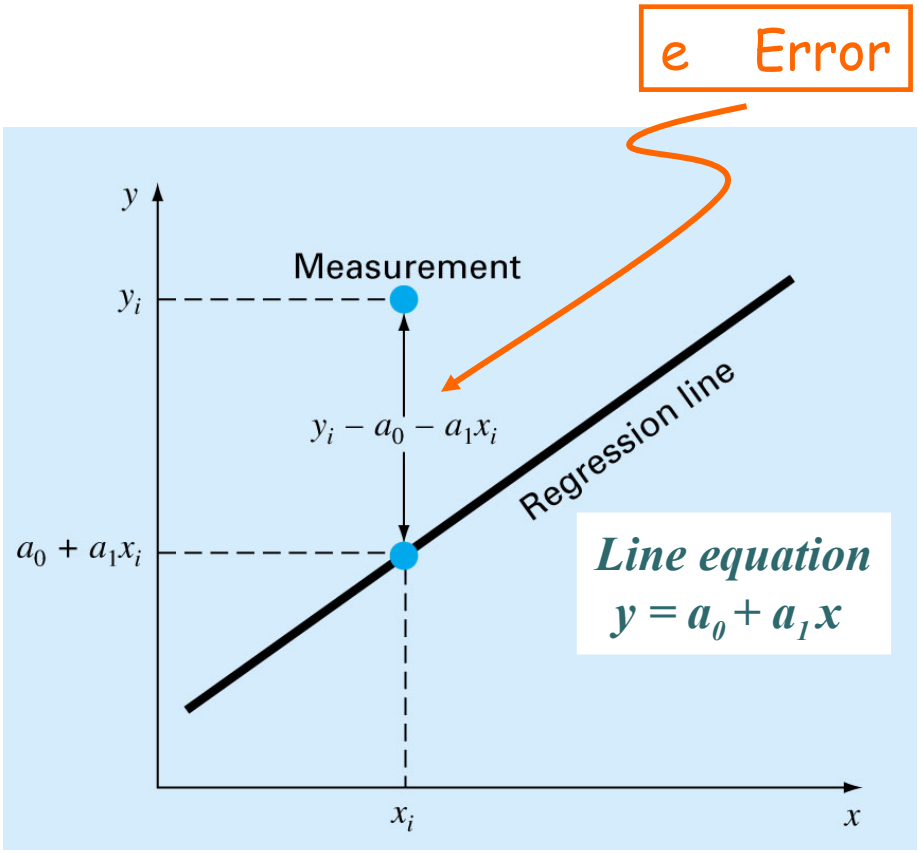
$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n).$$

- The mathematical expression for the straight line is

$$y = a_0 + a_1 x + e \qquad (1)$$

where a_0 and a_1 are coefficients representing the intercept and the slope, respectively. The error, or residual, between the model and the observations, which can be represented by rearranging Eq. (1) as

$$e = y - a_0 - a_1 x \qquad (2)$$



Residuals

- The residuals are the differences between data values and corresponding calculated fit data. The total error

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i) \qquad (3)$$

where *n* = total number of points.

- To minimize the sum of the squares of the residuals between the measured *y* and the *y* calculated with the linear model

$$S_r = \sum_{i=1}^{n} {e_i}^2 = \sum_{i=1}^{n} \left(y_{i,measured} - y_{i,model}\right)^2 = \sum_{i=1}^{n} (y_i - a_0 - a_1 x_i)^2 \qquad (4)$$

S_r is called the *sum of the square of the residuals*.

Least-Squares Fit of a Straight Line

- The technique for determining the values of a_0 and a_1 that minimize S_r is as following:
- To determine values for a_0 and a_1, Eq. (3) is differentiated with respect to each coefficient

$$\frac{dS_r}{da_0} = -2\sum (y_i - a_0 - a_1 x_i)$$

$$\frac{dS_r}{da_1} = -2\sum (y_i - a_0 - a_1 x_i) x_i$$

- Setting these derivatives equal to zero will result in a minimum S_r. If this is done, the equations can be expressed as

$$0 = \sum y_i - \sum a_0 - \sum a_1 x_i$$

$$0 = \sum y_i x_i - \sum a_0 x_i - \sum a_1 {x_i}^2$$

- Note that $\sum a_0 = n a_0$.we can express the equations as a set of two simultaneous linear equations with two unknowns (a_0 and a_1):

$$n a_0 + \left(\sum x_i\right) a_1 = \sum y_i$$

$$\left(\sum x_i\right) a_0 + \left(\sum x_i^2\right) a_1 = \sum x_i y_i$$

These are called the ***normal equations***. They can be solved simultaneously

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2} \qquad (5)$$

$$a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n} \qquad (6)$$

Error of Least Square Regression

- The error associated with the least-square regression can be calculated using two main terms:

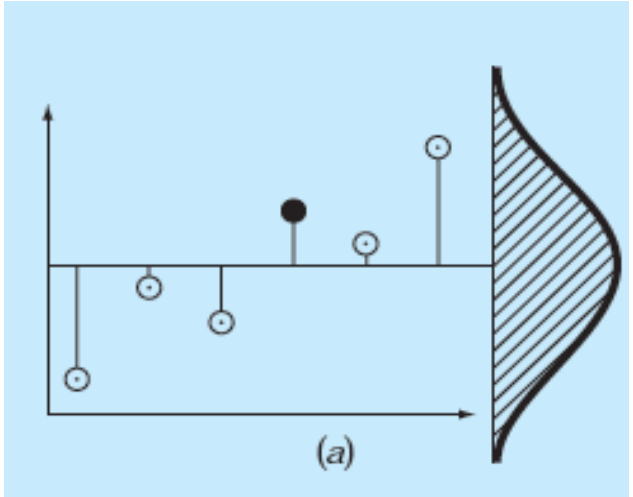
1. **The standard deviation** it's the spread *around the mean*, which represented by:

$$S_y = \sqrt{\frac{S_t}{n-1}}$$

where S_t is the total sum of the squares of the residuals between the data points and the mean

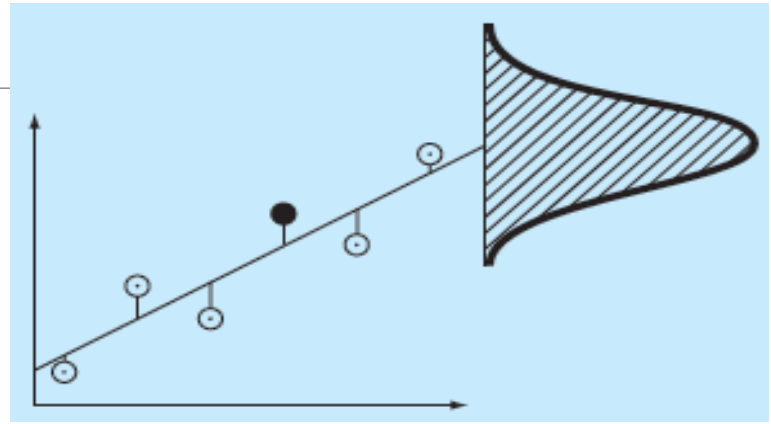
$$S_t = \sum (y_i - \overline{y_{model}})^2$$

S_y represents the square of the discrepancy between the data and a single estimate of the measure of central tendency.



2. **<u>The standard error (or root-mean square error)</u>** quantifies the spread *around the regression line*

$$S_{y/x} = \sqrt{\frac{S_r}{n-2}} \qquad (7)$$



where

$$S_r = \sum_{i=1}^{n} (y_i - y_{model})^2 \qquad (8)$$

The subscript notation "*y/x*" designates that the error is for a predicted value of *y* corresponding to a particular value of *x.* Also, notice that we now divide by *n* − 2 because two data-derived estimates—*a*0 and *a*1— were used to compute *Sr*; thus, we have lost two degrees of freedom.

3. The correlation coefficient:

$$r^2 = \frac{S_t - S_r}{S_t} \Rightarrow r^2 = \frac{S_t - S_r}{S_t} \qquad (9)$$

r^2 is the coefficient of determination. It indicates how many percent of data are explained by the linear model.

Summary

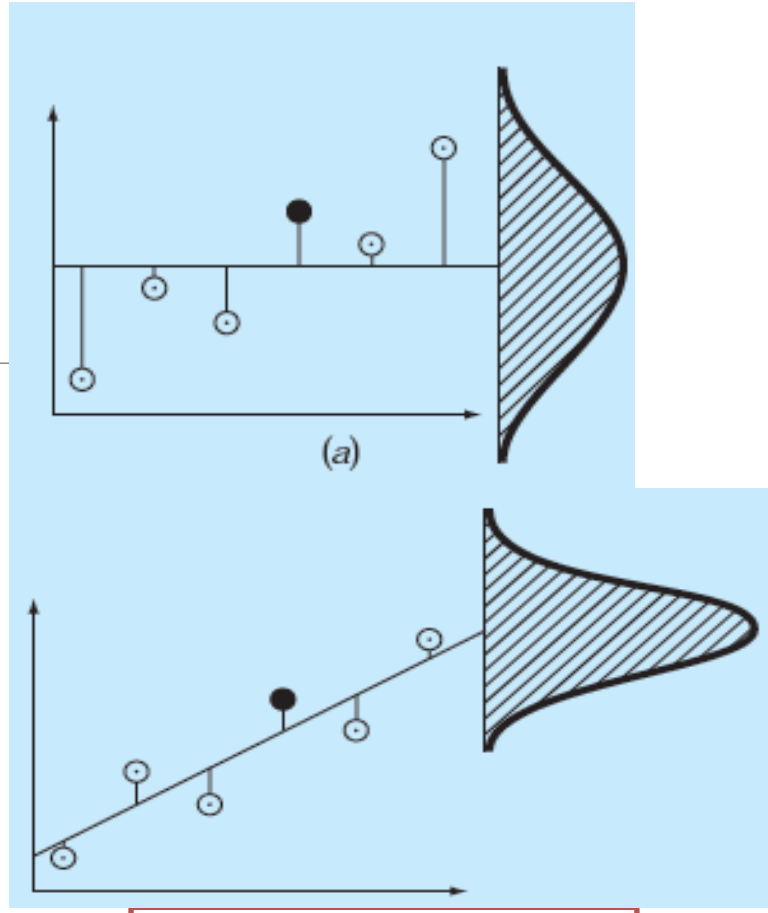
- S_r = **Sum of the squares of residuals around the regression line**
- S_t = **total sum of the squares around the mean**
- $(S_t - S_r)$ quantifies the improvement or error reduction due to describing data in terms of *a straight line* rather than as *an average value*.

➢ If $r^2 = 1$:

- then $S_r = 0$, which means that the **straight line is passing through all the data points and is a perfect fit.**

➢ If $r^2 = 0$:

- Then then $S_r = S_t$, that means that **regressing the data to a straight line does nothing to explain the data any further.**



The spread of data

a) *around the mean*
b) *around the best-fit line*

Notice the improvement in the error due to *linear regression*

Example (1)

Use least-squares regression to fit a straight line to the data given in following table.

X	6	7	11	15	17	21	23	29	29	37	39
Y	29	21	29	14	21	15	7	7	13	0	3

Along with the slope and the intercept, compute the standard error of the estimate and the correlation coefficient. Plot the data and the regression line.

Solution

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n}$$

$n = 11$

$\sum x_i y_i = 2380$

$\sum x_i^2 = 6262$

$\sum x_i = 234 \quad \bar{x} = 21.2727$

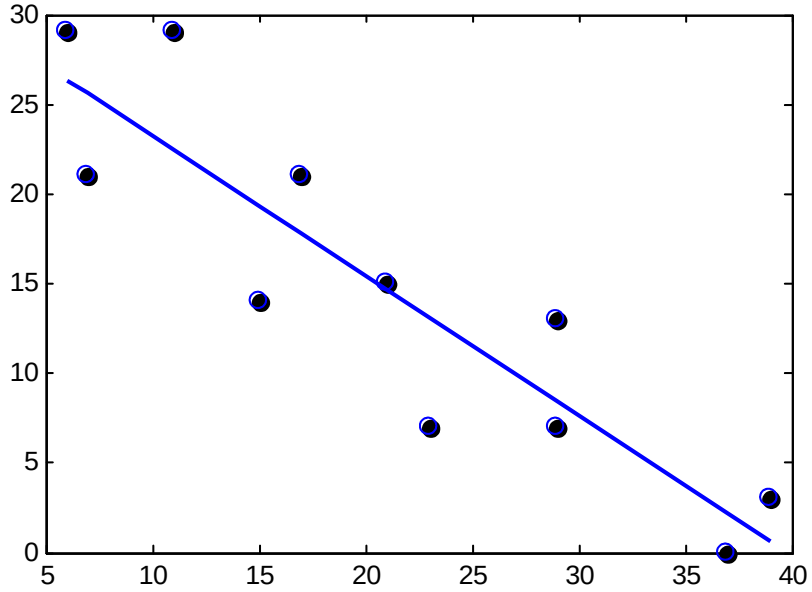
$\sum y_i = 159 \quad \bar{y} = 14.4545$

Using (5) and (6)

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2} = \frac{11 \times 2380 - 234 \times 159}{11 \times 6262 - (234)^2} = \quad -0.7805$$

$$a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n} = 14.4545 - \quad (-0.7805) \times 21.2727 = \quad 31.0589$$

However, the equation of the line y = 31.0589 -0.7805x,



$$y_{model} = 31.0598 - 0.7805x$$

x	6	7	11	15	17	21	23	29	29	37	39
y	29	21	29	14	21	15	7	7	13	0	3
y_{model}	26.3768	25.5963	22.4743	19.3523	17.7913	14.6693	13.1083	8.4253	8.4253	2.1813	0.6203

- To find the standard deviation,

$$\overline{y_{model}} = \frac{sum(y_{model})}{11} = 14.4564$$

$$S_t = \sum (y_i - \overline{y_{model}})^2 = (29 - 14.4564)^2 + (21 - 14.4564)^2 + \cdots = 962.7273$$

- The standard error

$$S_r = \sum_{i=1}^{n} (y_i - y_{model})^2 = (29 - 26.3768\ )^2 + (21 - 25.5963\ )^2 + \cdots = 180.3358$$

- The correlation coefficient:

$$r = \sqrt{\frac{S_t - S_r}{S_t}} = \sqrt{\frac{962.7273 - 180.3358}{962.7273}} = 0.9015$$

Nonlinear models using least squares

Linear regression provides a powerful technique for fitting a best line to data.

1. **Exponential model**

for a given points $(x_1, y_1), (x_2, y_2), .. (x_n, y_n)$, the exponential model:

$$y = ae^{bx} \quad (10)$$

Nonlinear models using least squares

Linear regression provides a powerful technique for fitting a best line to data.

1. Exponential model

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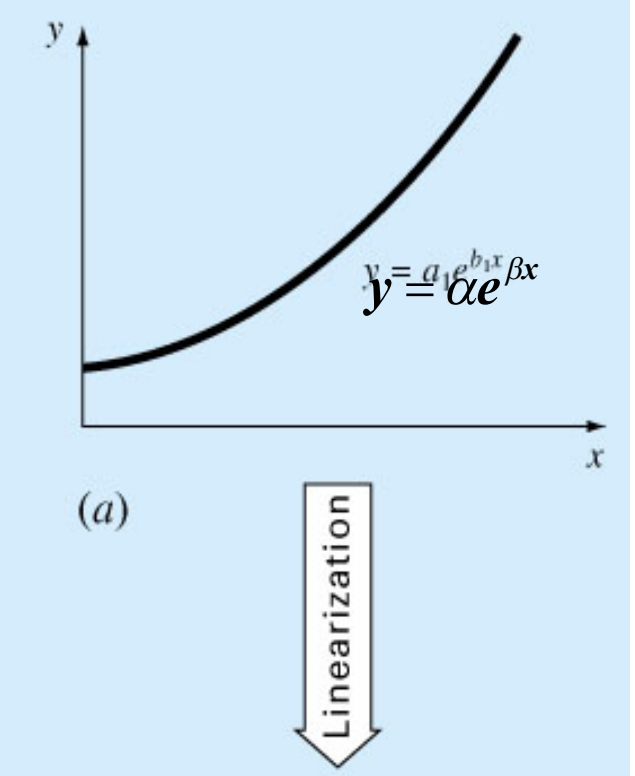
- The variables *a* and *b* are the constants of the exponential model.
- Eq. (10) can be linearized by taking its natural logarithm:

$$\ln y = ln\ a + bx\ ln\ e$$

- Because *ln e* = 1

$$\ln y = ln\ a + bx \quad (11)$$

- Thus a plot of *ln y* versus *x* yields a straight line with slope of *b* and an intercept of *ln a* .
- Now you can use the linear regression to evaluate the constant coefficient. Then transform them back to their original state.



Example (2)

Fit the following data using an exponential model

x	1	2	3	4
Y	0.5	1.7	3.4	5.7

Solution $\ln y = ln\, a + bx$

Table (1)

x	y	ln(y)	$x_i ln y_i$	x_i^2
1	0.5	-0.6931	-0.6931	1
2	1.7	0.5306	1.0613	4
3	3.4	1.2238	3.6713	9
4	5.7	1.7405	6.9619	16
$\sum x_i = 10$		$\sum (ln\, y_i) = 2.8017$	$\sum x_i ln y_i = 11.0013$	$\sum x_i^2 = 30$

All the required data are calculated in table (1).

The mean of *x* is $\bar{x}$ = 2.5000

The mean of *ln(y)* is $\overline{\ln(y)}$ = 0.7004

Using Eqs (5) and (6)

$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2} = \frac{4 \times 11.0013 - 10 \times 2.8017}{4 \times 30 - (10)^2} = 0.7994$$

$$a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n} = \frac{2.8017}{4} - 0.7994 \times \frac{10}{4} = -1.2981$$

- $a_0 = \ln(a) \implies a = e^{a_0} = e^{-1.2981} = 0.2731$
- $b = a_1 = 0.7994$

However, the exponential model is

$$y = 0.2731e^{0.7994\,x}$$

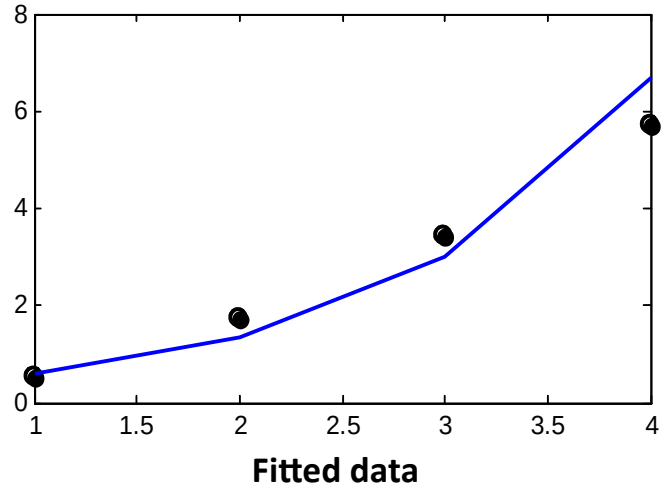
The errors are S_t = 15.2973, S_r = 1.2556 and r = 0.9581.

Note that S_r is v. small and r ≈ 1 , which means that the model gives good fitting.

Table (2)

A comparison between our model and the original data

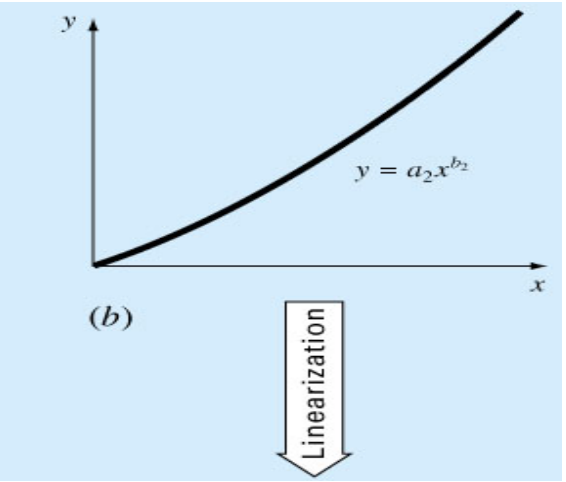
x	Y (Original)	$y = 0.2731e^{0.7994\,x}$ (model)
1	0.5	0.6073
2	1.7	1.3508
3	3.4	3.0046
4	5.7	6.6827



(2) Simple power model

The points $(x_1, y_1), (x_2, y_2), .. (x_n, y_n)$ are fitted to the model:

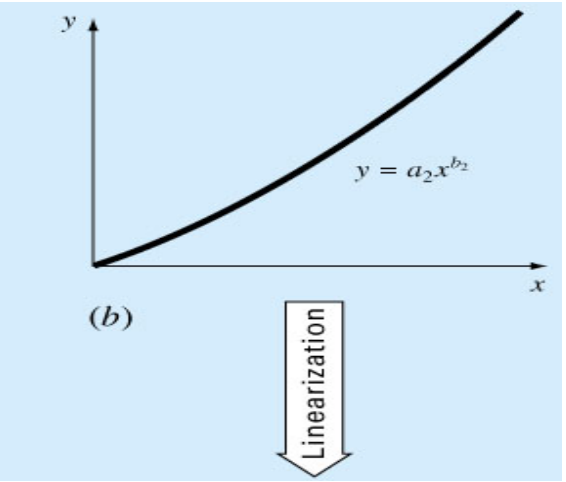
$$y = ax^b \quad (12)$$



(2) Simple power model

The points $(x_1, y_1), (x_2, y_2), .. (x_n, y_n)$ are fitted to the model:

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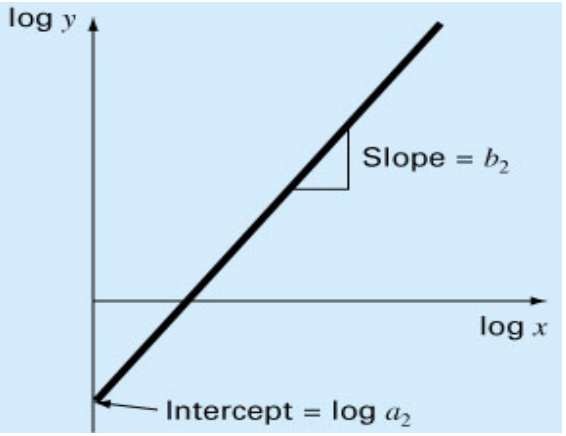
This equation can be linearized by taking its logarithm:

$$\log y = log\ a + b\ log\ x \quad (13)$$

It can be linearized by considering **$\log y = y_f$, $\log a = a_0$ and $\log x = x_f$**

to get $y_f = a_0 + a_1 x_f$

This equation is solved by following the same steps as it has been done in the previous section.



Example (3)

Repeat example (2) using simple-power model.

Solution

X	y	$y_f = \log(y)$	$x_f = \log(x_i)$	x_{fi}^2
1	0.5	-0.3010	0	0
2	1.7	0.2304	0.3010	0.0906
3	3.4	0.5315	0.4771	0.2276
4	5.7	0.7559	0.6021	0.3625
		$\sum(y_f) = 1.2168$	$\sum x_f = 1.3802$	$\sum x_{fi}^2 = 0.6807$

Table (3)

$$\sum x_{fi}y_{fi} = 0.7780 \qquad \overline{x_f} = 0.3451, \qquad \overline{y_f} = 0.3042$$

Using Eqs (5) and (6)

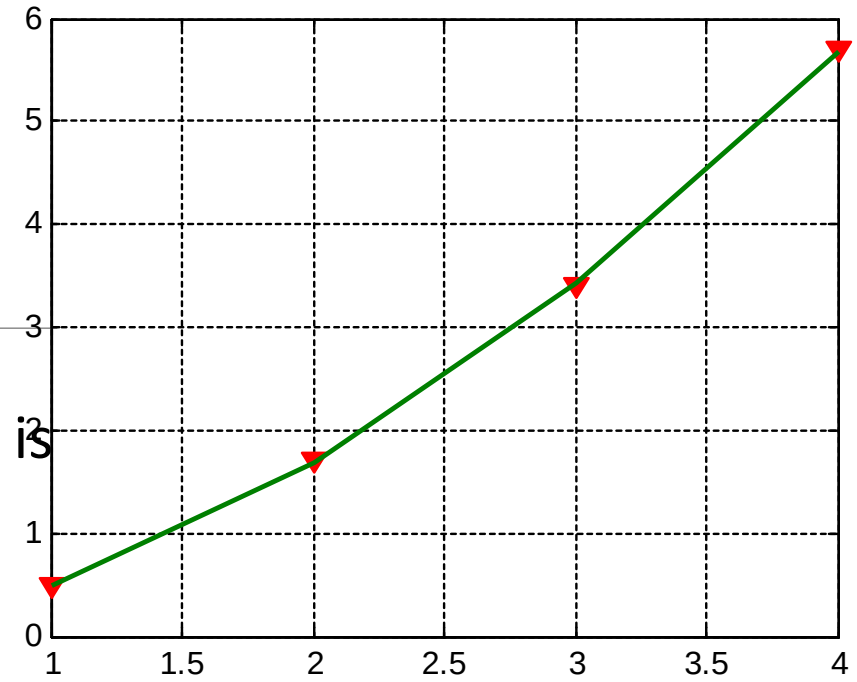
$$a_1 = \frac{n\sum x_{fi}y_{fi} - \sum x_{fi}\sum y_{fi}}{n\sum x_{fi}^2 - \left(\sum x_{fi}\right)^2} = \frac{4 \times 0.7780 - 1.3802 \times 1.2168}{4 \times 0.6807 - (1.3802)^2} = 1.7516$$

$$a_0 = \frac{\sum y_{fi}}{n} - a_1\frac{\sum x_{fi}}{n} = \frac{1.2168}{4} - 1.7516 \times \frac{1.3802}{4} = -0.3002$$

We have

$b = a_1$ and $\log a = a_0$ ➔,

a = 0.5010.



By substituting in (12) we get However, the power model is

$$y = 0.5010\, x^{1.7516}$$

- The calculated errors of our model are: S_t = 15.2675, S_r = 0.0016 and r = 0.9999.
- You can see that S_r is mostly zero and r ≈ 1 which refers to the good fitting.

- Tables (4) and (5) show comparison between the power model (Example (1)) and exponential model (example (2)).
- It can be seen, r of the power model is closer to 1 than of the exponential model . That means the power model gives mush better fitting than the exponential model.

Table (4)

x	y (Original)	$y = 0.501x^{0.7607}$ (Power model)	$y = 0.2731e^{0.7994\,x}$ (Exponential model)
1	0.5	0.5010	0.6073
2	1.7	1.6869	1.3508
3	3.4	3.4318	3.0046
4	5.7	5.6801	6.6827

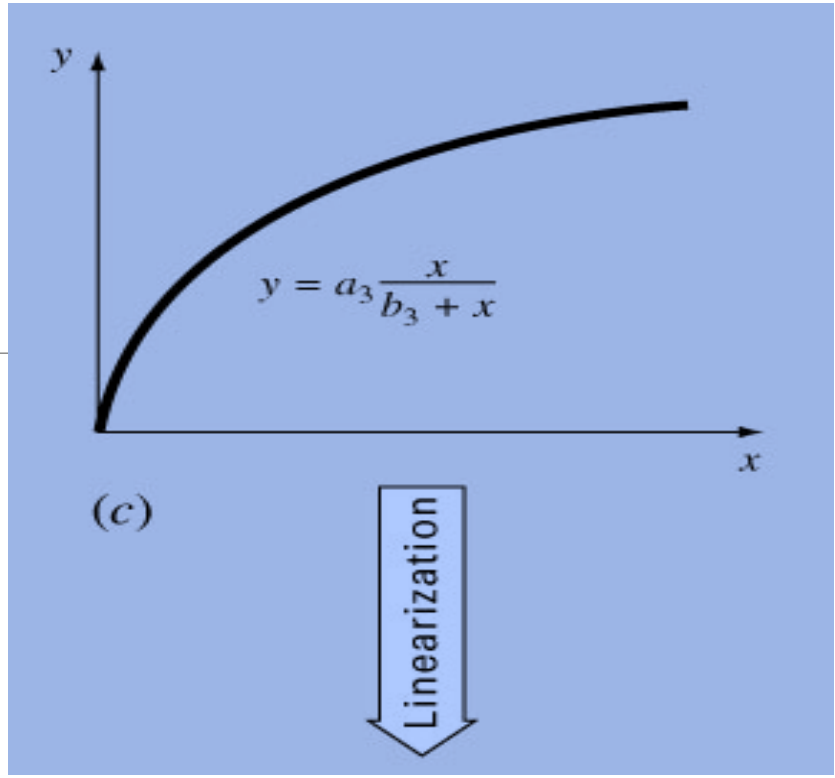
Table (5)

Errors	S_t	S_r	r
Power model	15.2675	0.0016	0.9999
Exponential model	3.3265	0.1313	0.9801

(3) Saturation-growth-rate model

The points $(x_1, y_1), (x_2, y_2), .. (x_n, y_n)$ are fitted to the model:

$$y = a\frac{x}{b + x} \qquad (14)$$



(3) Saturation-growth-rate model

The points $(x_1, y_1), (x_2, y_2), .. (x_n, y_n)$ are fitted to the model:

$$y = a\frac{x}{b + x} \qquad (14)$$

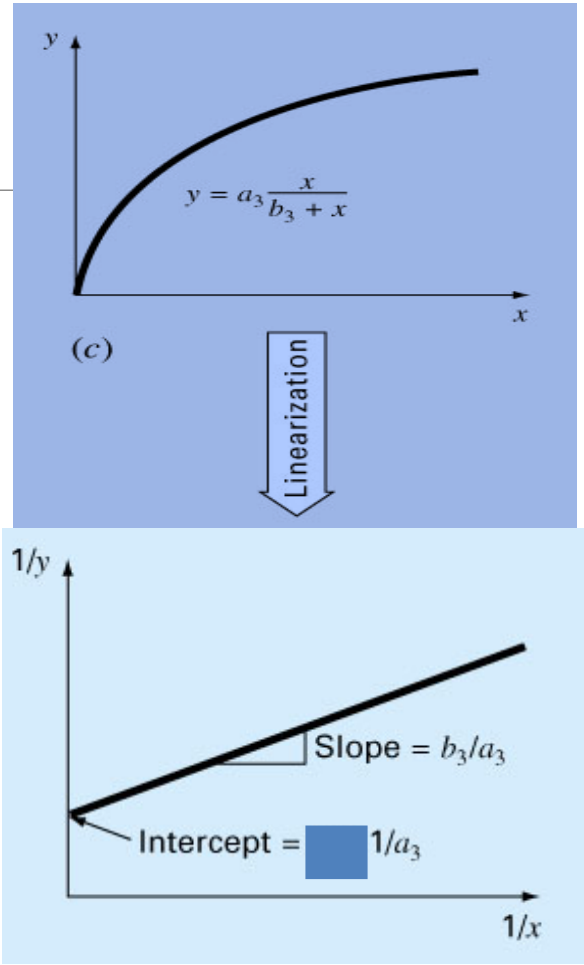
It is linearized by inverting it to give

$$\frac{1}{y} = \frac{b}{a} \times \frac{1}{x} + \frac{1}{a}$$

by considering $1/y = y_f$, $1/x = x_f$, $b/a = a_0$ and $1/a = a_0$ to get

$$y_f = a_0 + a_1 x_f$$

=This equation is solved by following the same steps as it has been done in the previous section



The data linearization of different types of functions

Function, $y = f(x)$	Linearized form, $Y = Ax + B$	Change of variable(s) and constants
$y = \frac{A}{x} + B$	$y = A\frac{1}{x} + B$	$X = \frac{1}{x}, Y = y$
$y = \frac{D}{x + C}$	$y + \frac{-1}{C}(xy) + \frac{D}{C}$	$X = xy, Y = y$
		$C = \frac{-1}{A}, D = \frac{-B}{A}$
$y = \frac{1}{Ax + B}$	$\frac{1}{y} = Ax + B$	$X = x, Y = \frac{1}{y}$
$y = \frac{x}{Ax + B}$	$\frac{1}{y} = A\frac{1}{x} + B$	$X = \frac{1}{x}, Y = \frac{1}{y}$
$y = A\ln(x) + B$	$y = A\ln(x) + B$	$X = \ln(x), Y = y$
$y = Ce^{Ax}$	$\ln(y) = Ax + \ln(C)$	$X = x, Y = \ln(y)$
		$C = e^B$
$y = Cx^A$	$\ln(y) = A\ln(x) + \ln(C)$	$X = \ln(x), Y = \ln(y)$
		$C = e^B$
$y = (Ax + B)^{-2}$	$y^{-1/2} = Ax + B$	$X = x, Y = y^{-1/2}$
$y = Cxe^{-Dx}$	$\ln\left(\frac{y}{x}\right) = -Dx + \ln(C)$	$X = x, Y = \ln\left(\frac{y}{x}\right)$
		$C = e^B, D = -A$
$y = \frac{L}{1 + Ce^{Ax}}$	$\ln\left(\frac{L}{y} - 1\right) = Ax + \ln(C)$	$X = x, Y = \ln\left(\frac{L}{y} - 1\right)$
		$C = e^B$ and L is a constant that must be given